Chapter 3 Teacher Key

Ramp It Up 2.0 Key

REQUIRED lab. There is no single correct centimeter table. Students record their own trials.

Look for: - one ball - vertical ramp height in centimeters for 1, 2, and 3 books - one unrecorded calibration roll - three recorded trials at each height (nine recorded trials) - an average column - a graph of ramp height (cm) against average cup movement (cm) with a visible grid - the CONNECT TO SPEED and CONNECT TO ENERGY reasoning steps (not collapsed into ramp height = speed) - no speed number and no meters per second - the 30 cm challenge judged by closeness without going past

Expected claim: When the same object moves faster, it has greater energy of motion. Acceptable wording: "more energy of motion", "greater energy". Evidence should contain TWO actual average cup-movement values from the student's own table. Reasoning should connect a larger average to a greater change. Misconception to catch: ramp height IS the speed, or that a speed number was measured.

A steeper ramp should usually move the cup farther. If it does not, help the student check the fair test before changing the science claim.

Answer Key — Drop It

- **Source alignment:** AP 4.2 (Drop It, Day 2); Reader Ch.3 pp. 13–16; TG L4. ADAPTED (fixed cm heights + cm crater width; depth qualitative).

Item	Correct/ expected response	Acceptable alternatives	Reasoning	Common misconception	Scoring note	What NOT to penalize
PREDICT	a guess (e.g., "90 cm makes the biggest crater")	any prediction	prediction = testable guess	wrong prediction = failure	1 pt	a "wrong" prediction
FAIR TEST (change/keep/measure)	change drop height; keep ball/sand tray/surface/drop	paraphrases	fair-test isolation	thinking height = speed (it's a cause, speed is inferred)	3 pts	informal phrasing

Item	Correct/ expected response	Acceptable alternatives	Reasoning	Common misconception	Scoring note	What NOT to penalize
	technique; measure crater width (cm)					
DATA TABLE	crater width increases with drop height; averages computed	any internally consistent data	repeated trials + average	recording "energy" in a data cell	6 pts	noisy crater data
OBSERVATION	60 cm (or 90 cm) caused the greatest change; depth note (medium/deep)	any correct height + qualitative depth	higher → faster → bigger crater	—	2 pts	depth wording
EXPLANATION	higher drop → faster fall (inferred) → bigger crater (measured) → greater energy of motion (inferred)	claim + measured change + inference	change = evidence of energy	claiming we measured energy or speed	3 pts	informal phrasing

Notes

- Crater depth is qualitative only (shallow / medium / deep) — do not require millimeter precision.
- Students must NOT claim they measured the ball's speed or energy. The "faster" determination is an inference from drop height.
- Fixed heights (30/60/90 cm) replace the official knee/waist heights for a fairer test (see the matching Chapter 3 resource).

Chapter 3 Practice Key

Not graded. OPTIONAL.

1. The farther cup move is the evidence. Accept: a bigger change, or the cup moved farther. Do not accept a

speed number.
2. A different ball could be heavier or lighter, so the test would compare different objects instead of one object at two speeds.
3. Measured: ramp height in centimeters, or how far the cup moved in centimeters. Compared: a lower setup and a steeper setup (for example, 1 book and 3 books) for the same ball.

Reader pp. 13–16. Teacher Guide Lessons 4 and 5.

Chapter 3 Quiz Key

8 items · 12 points. Score: __ **/ 12 =** ____ %. FLEX. Friday 2026-10-23 or makeup within 2 school days. Not a first lesson.

1. (1 point) distance. Reader p. 13.
2. (1 point) A. one ball at two speeds.
3. (1 point) ramp height. The ball stayed the same.
4. (1 point) A. ramp height (cm) and average cup movement (cm).

Scoring for the four 2-point items (3, 4, 7, 8):

3. (2 points) What shows that the faster ball had more energy of motion? 2 = names a greater change (a farther cup move, a higher bounce, or a louder sound) with no speed number. 1 = names a change but not a greater one. 0 = incorrect, or gives a speed number.

4. (2 points) In Ramp It Up 2.0, what did we measure in centimeters? Name both things. 2 = names both: ramp height in centimeters AND cup movement in centimeters. 1 = names only one of the two. 0 = neither, or gives a speed number.

5. (2 points) The same ball rolls from a steeper ramp. What greater change do you look for? 2 = names a greater change for the same ball (the cup or bucket moving farther). 1 = names a change but not a greater one. 0 = incorrect.

6. (2 points) How are energy and speed related? 2 = same object, and greater speed relates to greater energy of motion, with a greater change as the evidence. 1 = states the relationship for the same object but omits the evidence, or states evidence without the relationship. 0 = a mass-based comparison, a speed number, or no same object.

Do not accept kinetic, potential, mechanical, or a speed in meters per second.